Capstone Build

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected capstone-build record.

Explain stages, review a mismatch, audit generalisation and design a revealing integration case.

Keep one habit: Keep build version, source facts, missing values, observed outputs and unresolved claims together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Capstone Design

Safety Boundaries

Privacy by Design

W1 · Explain the build stages

> Connect input, eligible source, validation and displayed result.

Your evidence record

- Room W2 review: request B, approval B, register C capacity 10; same-room support fails, needs_review/null claims. Main investigation E2: 3 cups observed 4 days, losses/initial volumes absent. E3 adds losses 1, 2, 3 mLbut no initial volumes, so half status unknown. Program validates a narrow structured record and formats supplied evidence; it does not parse arbitrary text or authenticate measurement.

  Seed assessment uses separate facts.

- Reference for this exercise: open this week's student lesson, "Capstone Build", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Review the room mismatch

> Keep this repeated case as integration practice.

Your evidence record

- Room W2 review: request B, approval B, register C capacity 10; same-room support fails, needs_review/null claims. Main investigation E2: 3 cups observed 4 days, losses/initial volumes absent. E3 adds losses 1, 2, 3 mLbut no initial volumes, so half status unknown. Program validates a narrow structured record and formats supplied evidence; it does not parse arbitrary text or authenticate measurement.

  Seed assessment uses separate facts.

- Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit a happy-path guarantee

> One supported path does not establish every failure boundary.

Your evidence record

- Room W2 review: request B, approval B, register C capacity 10; same-room support fails, needs_review/null claims. Main investigation E2: 3 cups observed 4 days, losses/initial volumes absent. E3 adds losses 1, 2, 3 mLbut no initial volumes, so half status unknown. Program validates a narrow structured record and formats supplied evidence; it does not parse arbitrary text or authenticate measurement.

  Seed assessment uses separate facts.

- A report omits this qualification: A correct happy path does not establish fallback, access control or side-effect safety. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing integration case

> Supply exact facts and predicted failure stage before trying it.

Your evidence record

- Room W2 review: request B, approval B, register C capacity 10; same-room support fails, needs_review/null claims. Main investigation E2: 3 cups observed 4 days, losses/initial volumes absent. E3 adds losses 1, 2, 3 mLbut no initial volumes, so half status unknown. Program validates a narrow structured record and formats supplied evidence; it does not parse arbitrary text or authenticate measurement.

  Seed assessment uses separate facts.

- Reference for this exercise: open this week’s student lesson, “Capstone Build”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Repair the new seed-investigation draft independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use only the independent experiment's stated procedure facts and missing results. No sprouting observations, soil comparison or later follow-up is supplied. Explain which claims may remain and which require new evidence; do not substitute the cup example's measurements into this seed task.

An experiment note records four fictional seeds observed for six days but supplies no growth results. A generated draft says all four sprouted and the method always works in any soil. Repair it.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Assemble interface, retrieval, prompt, model/tool, validation, **evidence** display, and fallback.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Capstone grounded assistant. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A pipeline retrieves passages, drafts and displays without checking citations.

What step is missing?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

The interface looks complete, but its answer button returns the same saved sentence for every question. What capability is still missing, and how can you expose it?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 34 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier zero-capacity room task

REFERENCE · SECTION 15

Additional practice · Earlier zero-capacity room task

> This retained exercise differs from the selected seed-investigation assessment.

Earlier task and response

- Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

Build a claim checker for a draft summary of a fictional class investigation.

Approved record E2: three cups were observed for four days. No measured water loss is provided. Draft says “three cups were observed for four days and all lost half their water”.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Build a claim checker for a draft summary of a fictional class investigation.
- Which labelled step matches this detail? Retain the supported procedure facts; state that water-loss results were not supplied.
- 1. Extract claims

- 2. Match **evidence**
- 3. Repair draft
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Repair draft Retain the supported procedure facts; state that water-loss results were not supplied. Read the picture in order.

  Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: E3 supplies losses of 1, 2 and 3 mL. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Report those individual losses, not “half” unless starting volumes are also available.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Report those individual losses, not “half” unless starting volumes are also available. Absolute losses alone do not establish each cup’s fractional loss.

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Capstone Design (ai-3/33). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Safety Boundaries (ai-3/31). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: If AB7 passes the pattern, may it access teacher-only material?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
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> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

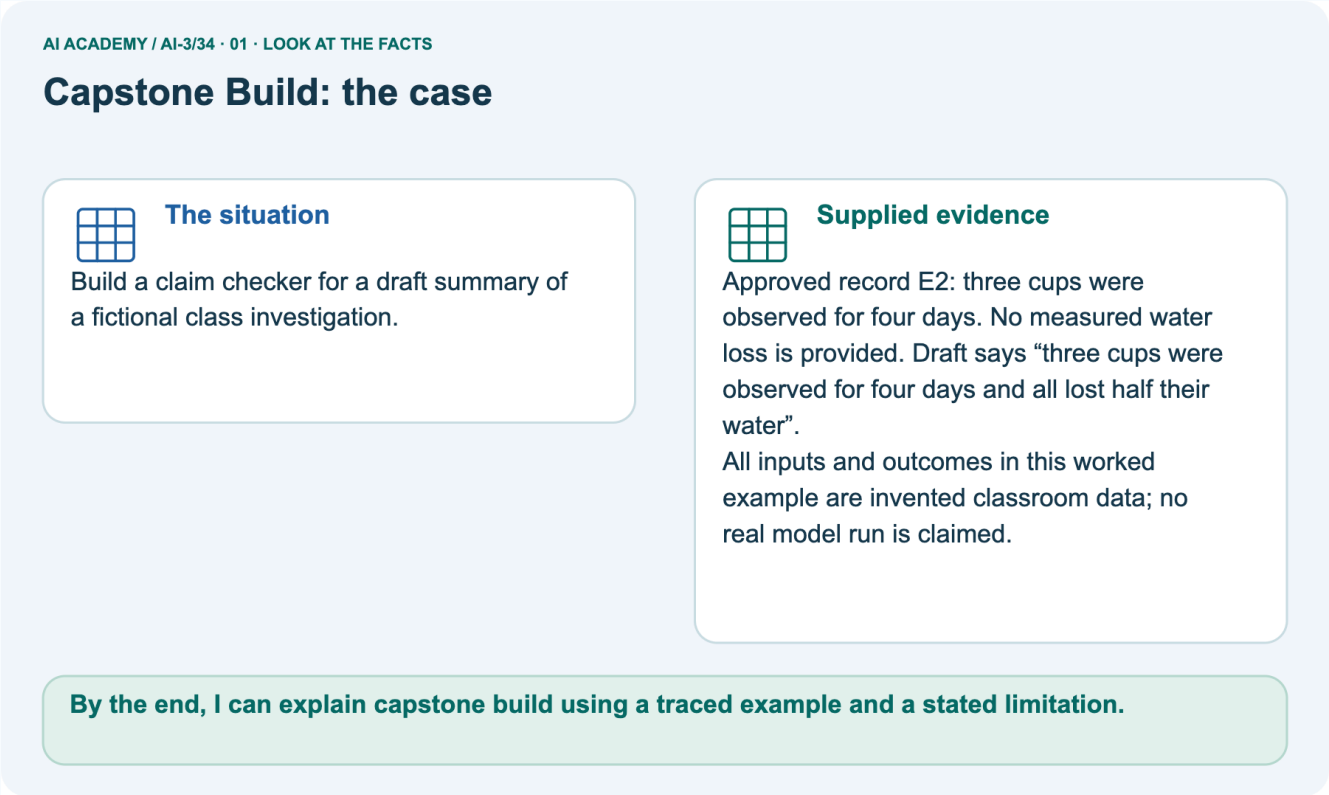
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Capstone Build: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

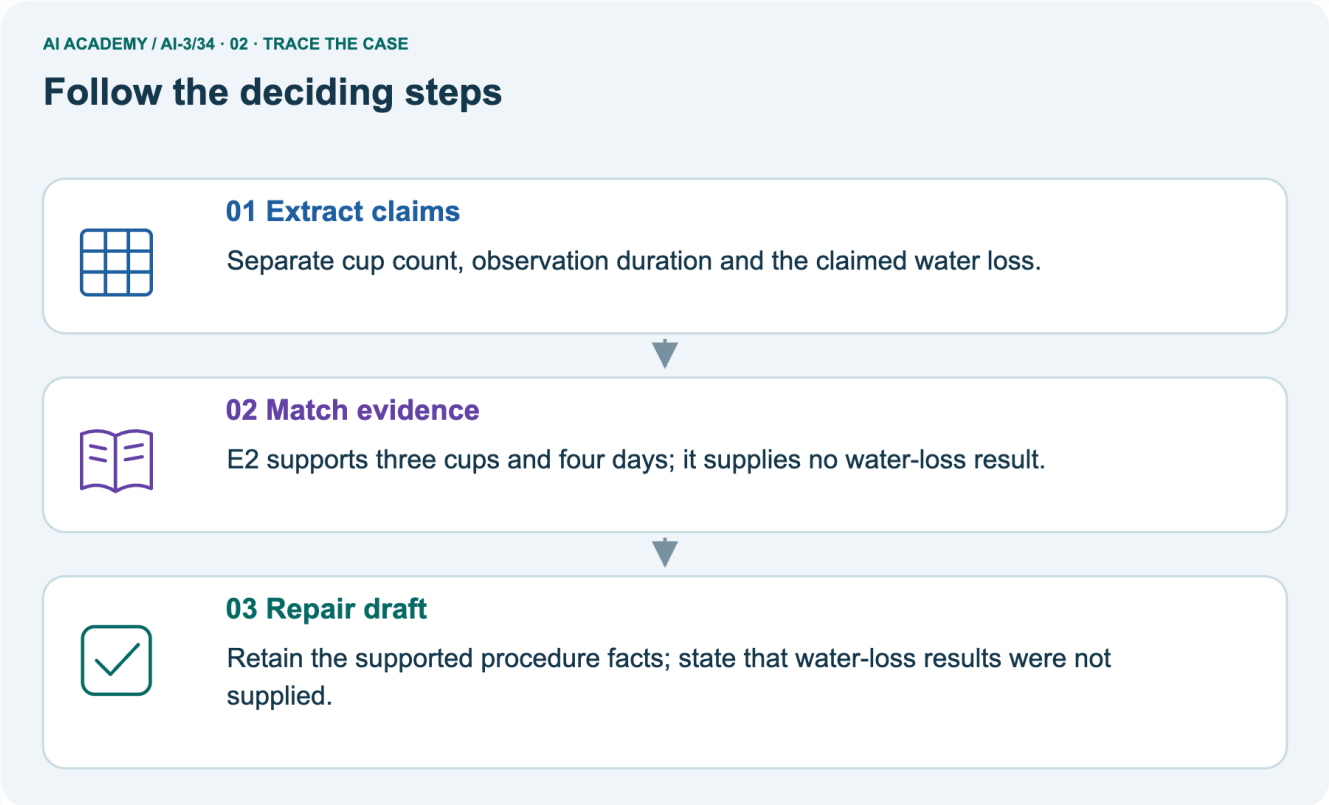
Build a claim checker for a draft summary of a fictional class investigation. Supplied facts: Approved record E2: three cups were observed for four days. No measured water loss is provided.

Draft says “three cups were observed for four days and all lost half their water”. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



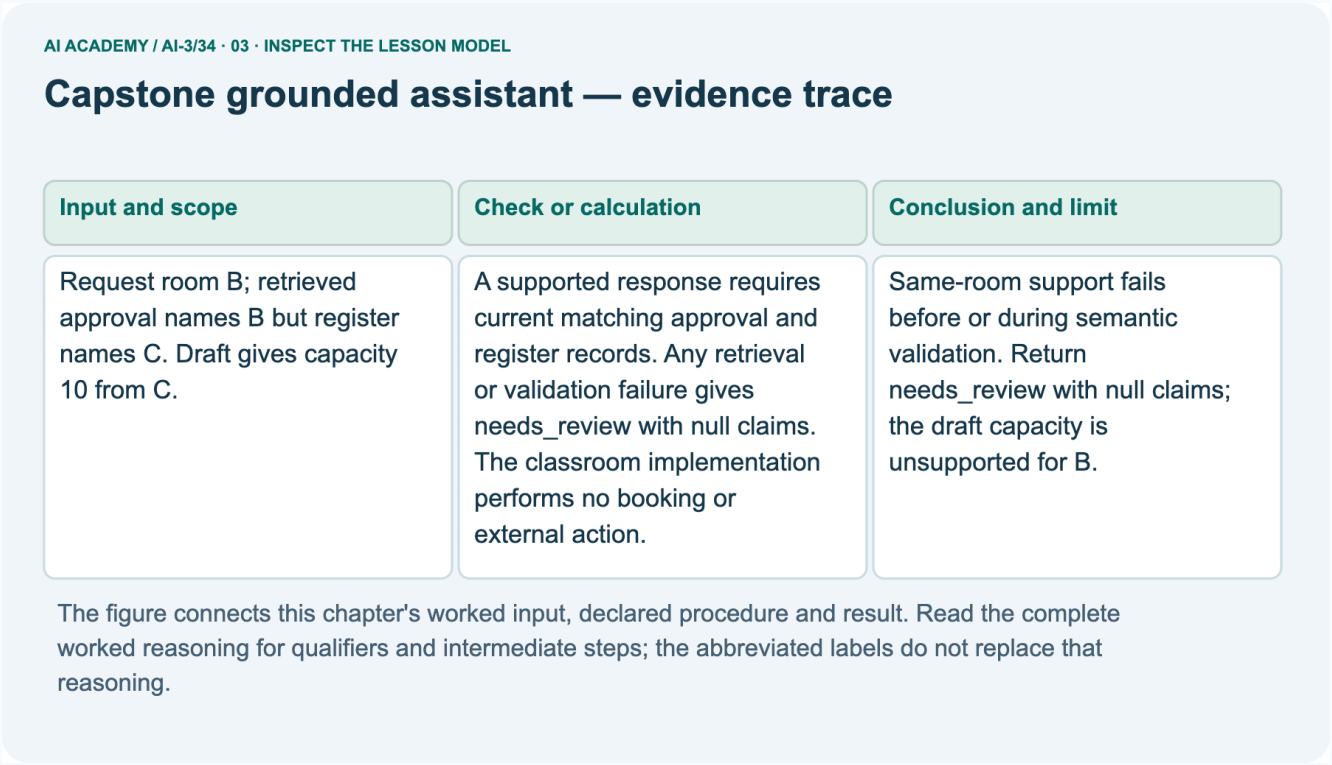
Follow the labelled picture with the matching explanation. Open the vector diagram

Extract claims: Separate cup count, **observation** duration and the claimed water loss. → Match **evidence**: E2 supports three cups and four days; it supplies no water-loss result. → Repair draft: Retain the supported procedure facts; state that water-loss results were not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone grounded assistant — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

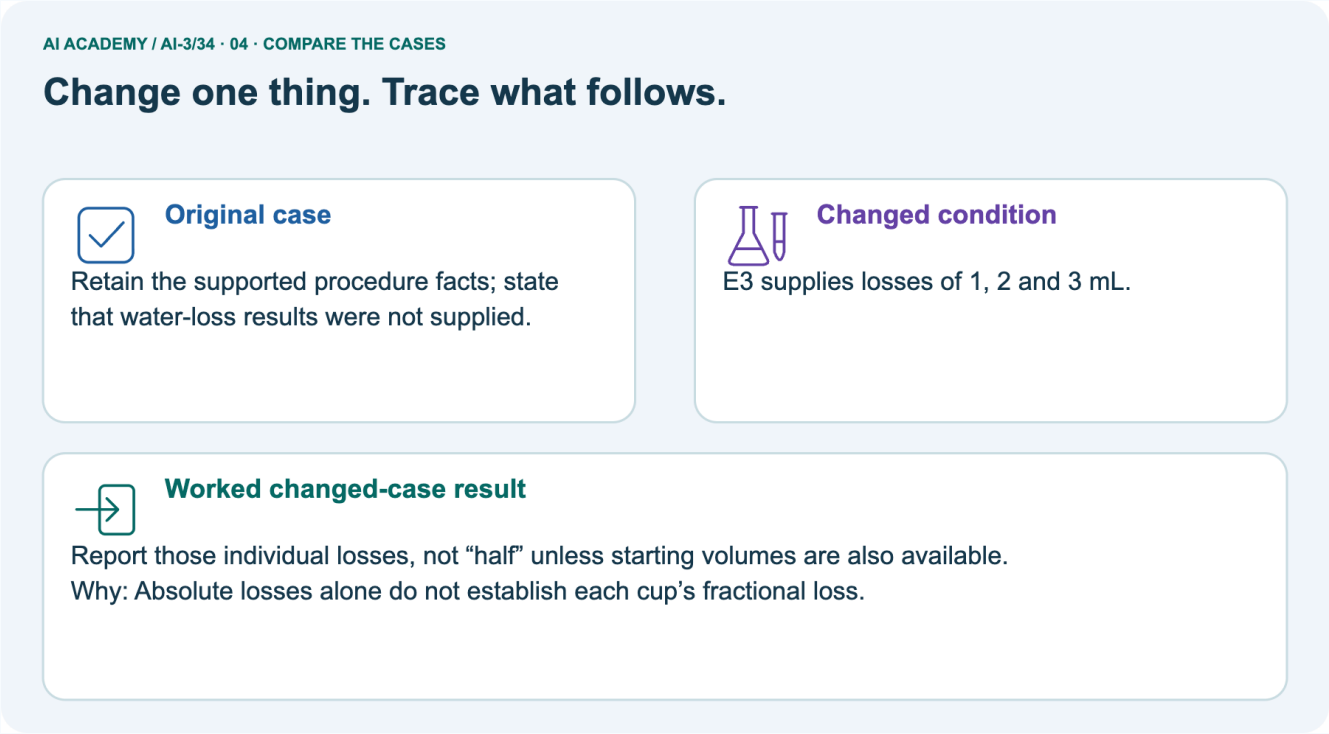
Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

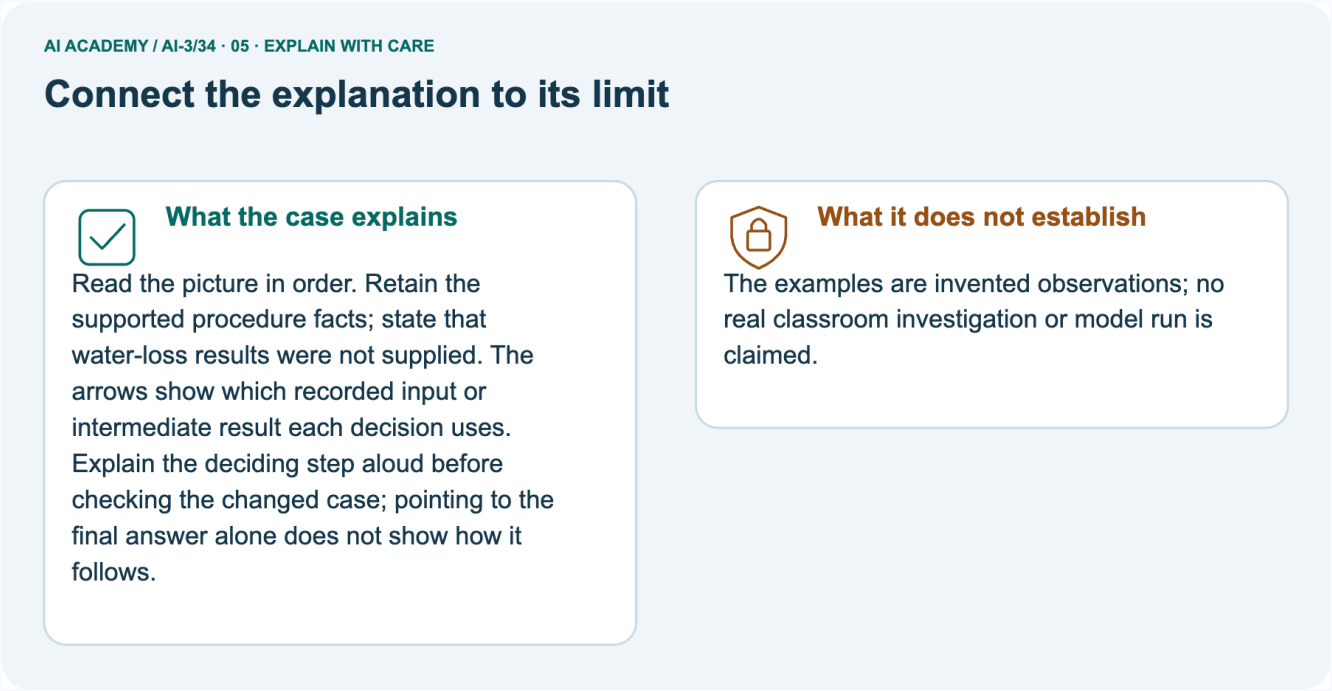
Original case: Retain the supported procedure facts; state that water-loss results were not supplied. Changed condition: E3 supplies losses of 1, 2 and 3 mL.

Expected result: Report those individual losses, not “half” unless starting volumes are also available. Reason: Absolute losses alone do not establish each cup’s fractional loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

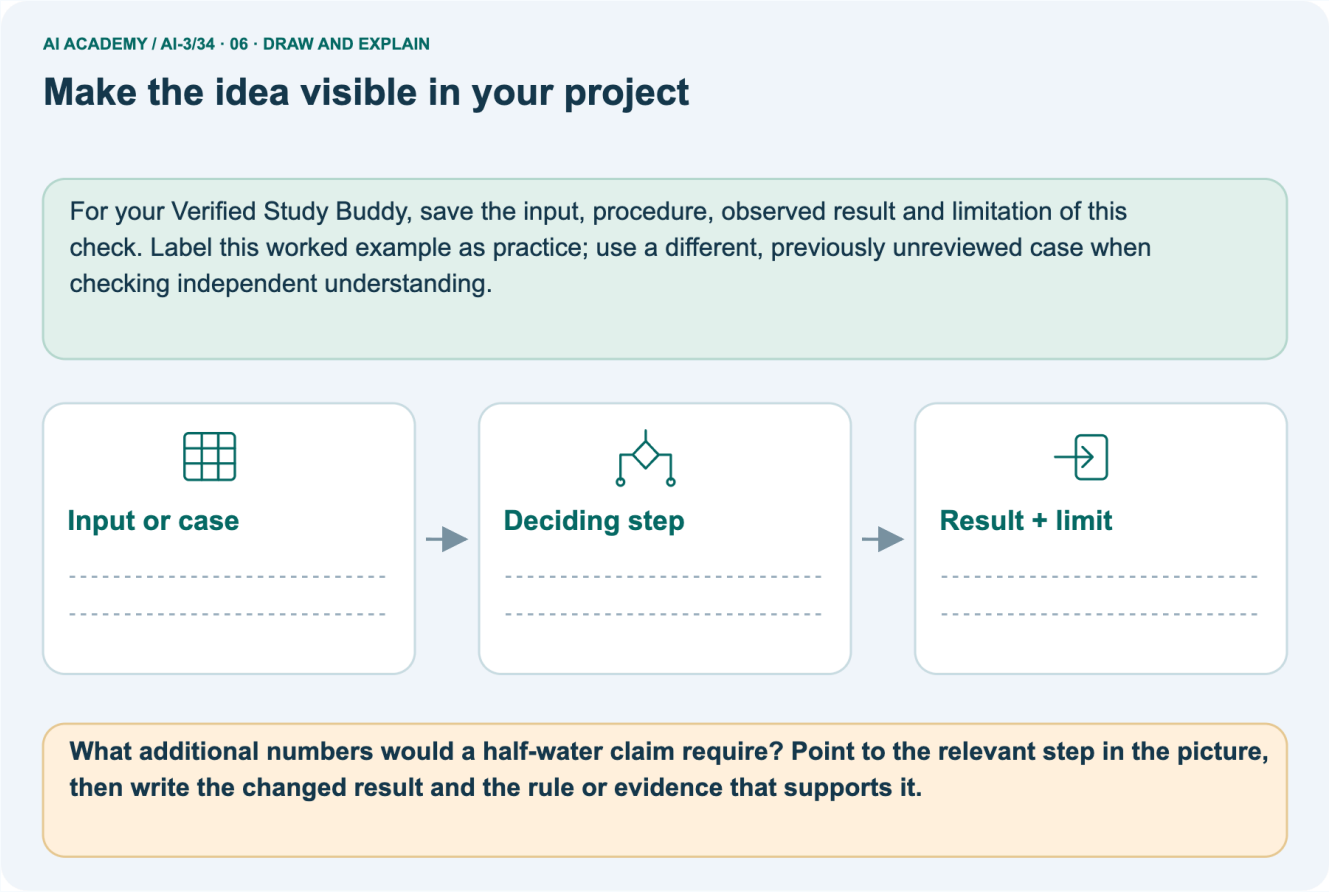
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The examples are invented **observation**s; no real classroom investigation or model run is claimed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Build: the case

02. Follow the deciding steps

03. Capstone grounded assistant — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.